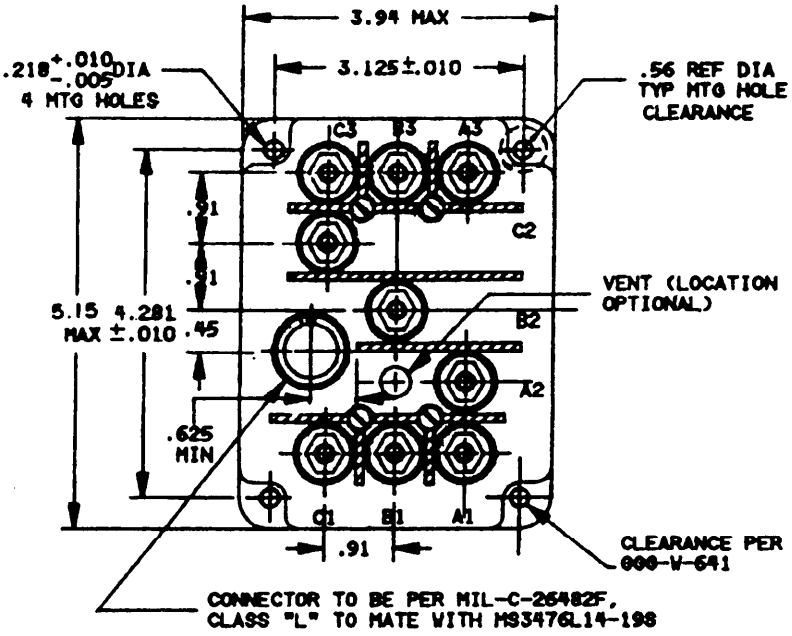
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 3PDT MAIN CONTACTS, 4PDT AUXILIARY CONTACTS, 135 AMPERES, 115/200 VOLTS, 400 HERTZ, GASKET SEALED.

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6106.



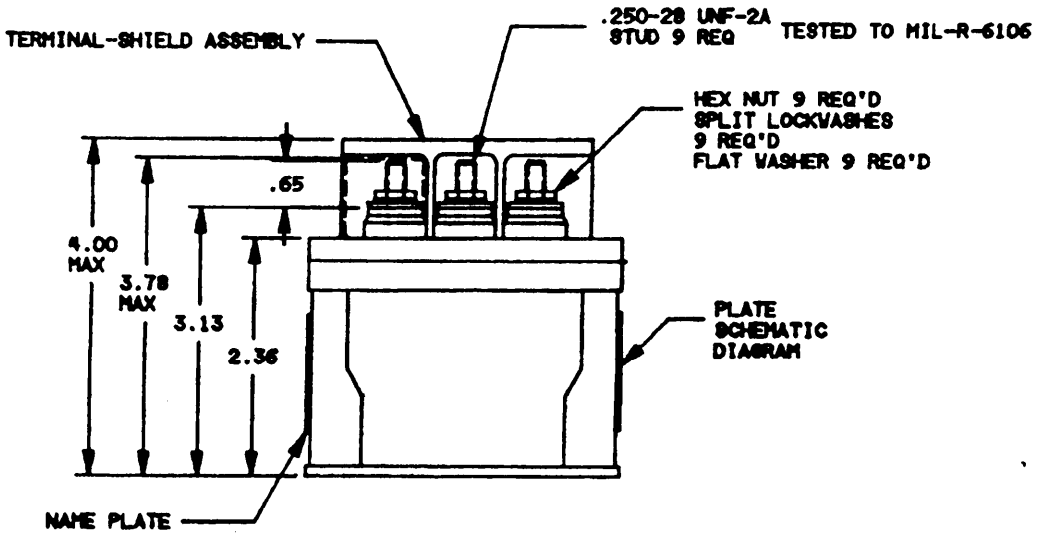
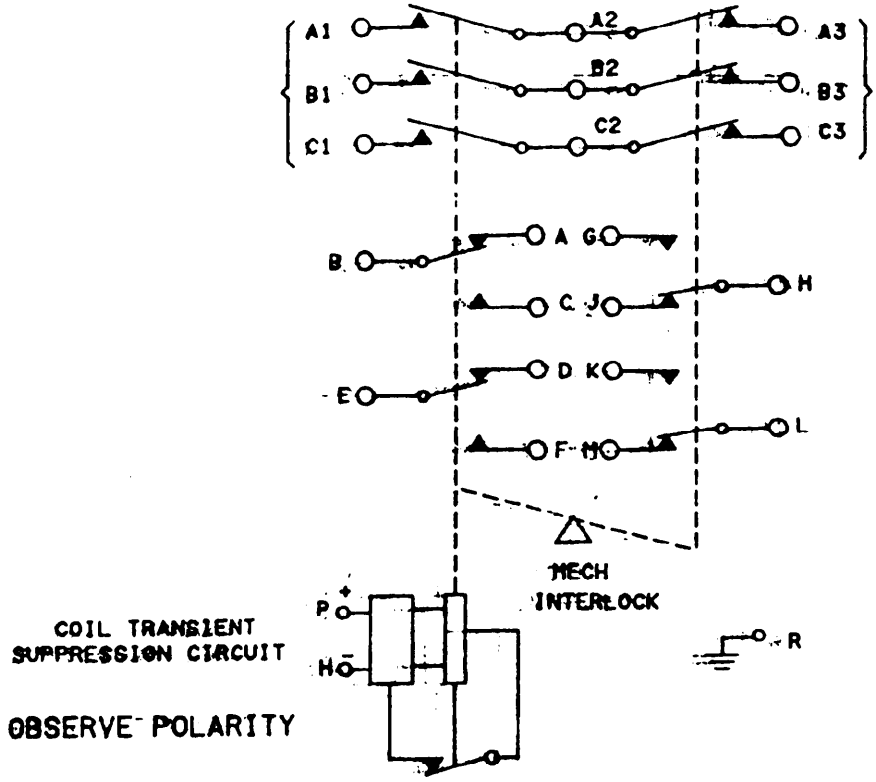


FIGURE 1. Relay outline drawing.

FSC 5945

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CIRCUIT DIAGRAM

INCHES	MM	INCHES	MM
.005	.12	2.36	59.9
.010	.25	3.125	79.37
.218	5.53	3.13	79.5
.250	6.35	3.78	96.0
.45	11.4	3.94	100.1
.56	14.2	4.00	101.6
.65	16.5	4.28	108.7
.91	23.1	5.15	130.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.

FIGURE 1. Relay outline drawing - Continued.

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REQUIREMENTS:

Coil data:

Coil data: See table I.

Duty cycle: Continuous.

Coil suppression: 42 V dc maximum for 5 seconds, shall meet the requirements of MIL-E-6051.

Operational data:

Operate time: See table I.

Release time: See table I.

Rated contact load: See table II.

Physical data:

Dimensions and configuration: See figure 1.

Weight (maximum): 3.15 pounds.

Terminal type: See figure 1.

Seal: Gasket sealed, vented.

Environmental characteristics:

Operating temperature: -55°C to -71°C.

Altitude: 50,000 feet, maximum.

Vibration: Per MIL-E-5400, curve IV, 5 to 500 hertz minimum.

Shock: 18 G's, minimum, 11 milliseconds duration.

Acceleration: 6 G's.

Electrical:

Insulation resistance (initial): 200 megohms.

After life or environmental tests: 100 megohms.

Contact voltage drop (initial): .175 volts maximum, .150 volts average.

After life test: .200 volts maximum, .175 volts average.

Overload current (main): 1,080 amperes.

Rupture current (main): 1,350 amperes.

Dielectric withstanding voltage (see level): See table III.

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TABLE I. Operating characteristics.

Dash no.	Coil data										Time ms (maximum) 2/		
	Nominal		Maximum			Pickup voltage (maximum)			Dropout voltage 1/	Hold Voltage	Coil voltage 18 V dc	28 V dc	30 V dc
	Volts	Frequency	Volts	Amperes		Δt 25°C	Hi-temp	Continuous currents			Operate	Release	Transfer
				In-rush	Continuous								
001	28	dc	30	5	1	15	18	22.5	1.0	7.0	50	35	10

1/ Over temperature range.
2/ With nominal coil voltage at 25°C.

TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	50/60 Hz	400 Hz	60 Hz	400 Hz	50/60 Hz	400 Hz	60 Hz
Resistive	50	---	---	5	5	135	---	5	---	135	---	5	---
Inductive	50	---	---	3	3	135	---	3	---	135	---	3	---
Motor	50	---	---	---	---	80	---	---	---	80	---	---	---
Lamp	---	---	---	2	2	---	---	2	---	---	---	2	---
Transfer load	10	---	---	---	---	135	---	---	---	135	---	---	---
Mechanical life reduced current	100	---	---	1.25	1.25	33.75	---	1.25	---	33.75	---	1.25	---
Intermediate current	50	13.5	13.5	Per MIL-R-6106									

1/ Absence of value indicates relay is not rated for 3 phase application.

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TABLE III. Dielectric strength at sea level.

Dielectric strength at sea level (2-5 seconds)				
	Test voltages			
	Initial		After life	
	28 V dc	115 Vac	28 V dc	115 V ac
Coil to case	1,250	---	1,000	---
Aux. contacts	1,250	1,500	1,000	1,125
All other points	N/A	1,800	N/A	1,350

Dielectric withstanding voltage (altitude): See table IV.

TABLE IV. Dielectric strength at altitude.

Dielectric strength at altitude (60 seconds)		
	Test voltages	
	28 V dc	115 V ac
Coil to case	500	---
Aux. contacts	500	500
All other points	N/A	700

Part number: M6106/42-001.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5945-F519)

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